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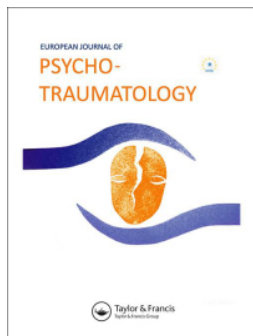


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CLINICAL RESEARCH ARTICLE



Polish adaptation and validation of the International Trauma Questionnaire (ITQ) for PTSD and cPTSD according to ICD-11 in non-clinical and clinical samples

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ABSTRACT

Background: The International Trauma Questionnaire (ITQ) is a validated measure supporting the diagnosis of ICD-11 post-traumatic stress disorder (PTSD) and complex PTSD (cPTSD). It has been translated into 35 languages, also into Polish. Due to the lack of information in the literature about its validation, new Polish version has been re-adapted and validated in the Polish non-clinical and clinical samples of the present study.

Objective: The study aimed (1) to adapt the ITQ for use in Polish, (2) to assess the construct validity of ICD-11 PTSD and cPTSD in the Polish ITQ, and (3) to examine the convergent and discriminant validity of the Polish ITQ.

Method: The study sample comprised of 452 adults: a non-clinical sample, $N = 314$, (completed the online version of the questionnaire) and a clinical sample, $N = 138$, (used a pen-and-paper version). Confirmatory Factor Analysis (CFA) and Latent Class Analysis (LCA) were performed to assess the structure of the ITQ.

Results: The CFA supported a two-factor second-order model as the best fit to the data. All factor loadings were statistically significant and reached a satisfactory level. Cronbach's alpha for ITQ ($\alpha = 0.887$), PTSD ($\alpha = 0.810$) and DSO ($\alpha = 0.875$) were good. The LPA revealed four classes: (1) a cPTSD class with elevated PTSD and DSO symptoms; (2) a PTSD class with elevated PTSD symptoms but low scores on the DSO; (3) a DSO class with elevated DSO symptoms but low scores on the PTSD; and (4) a low symptom class with low scores on all symptoms.

Conclusion: The amended Polish ITQ has good internal consistency and psychometric properties. The model is well fitted to differentiating PTSD/cPTSD diagnoses according to ICD-11 and is a reliable measure of support for the making of a PTSD/cPTSD diagnosis in the conduct of clinical interview.

Adaptación y validación Polaca del Cuestionario Internacional de Trauma (ITQ) para el TEPT y TEPTc según le CIE-11 en muestras no clínicas y clínicas

Antecedentes: El cuestionario Internacional de Trauma (ITQ por sus siglas en ingles) es una medida validada que respalda los diagnósticos de trastorno de estrés postraumático (TEPT) y TEPT complejo según la CIE-11. Ha sido traducido a 35 idiomas. También está disponible una traducción al polaco de la ITQ; sin embargo, debido a la falta de información en la literatura acerca de su validación, ahora ha sido re-adaptado y se ha validado una nueva versión en polaco en las muestras no clínicas y clínica polacas del presente estudio.

Objetivos: Este estudio tuvo como objetivos (1) Adaptar el ITQ para su uso en polaco, (2) Evaluar la validez del constructo del TEPT y TEPTc de la CIE-11 en el ITQ polaco, y (3) Examinar la validez convergente y discriminante del ITQ polaco.

Método: La muestra del estudio estuvo compuesta de 452 adultos: una muestra no clínica, $N = 314$, (que completaron la versión en línea del cuestionario) y una muestra clínica, $N = 138$, (que utilizó la versión de papel y lápiz). Se realizaron Análisis de Factor Confirmatorio (AFC) y Análisis de Clases Latente (ACL) para evaluar la estructura del ITQ.

Resultados: El AFC respaldó un modelo de segundo orden de dos factores como el que mejor se ajustaba a los datos. Todas las cargas factoriales fueron estadísticamente significativas y alcanzaron un nivel satisfactorio. El alfa de Cronbach para el ITQ ($\alpha = 0.887$), TEPT ($\alpha = 0.810$) y DSO ($\alpha = 0.875$) fueron buenos. El PCL reveló cuatro clases: (1) una clase de TEPTc con síntomas elevados de TEPT y DSO; (2) una clase de TEPT con síntomas elevados de TEPT pero con puntuaciones bajas en el DSO; (3) una clase DSO con síntomas elevados de DSO pero con puntuaciones bajas en TEPT; y (4) una clase de

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PALABRAS CLAVE

TEPT complejo; TEPTc; Cuestionario Internacional de Trauma; ITQ; CIE-11; prevalencia; trauma complejo

HIGHLIGHTS

- This study is the first psychometric validation of the Polish adaptation of the ITQ questionnaire with a clinical and non-clinical sample.
- The study presents the prevalence of ICD-11 PTSD and cPTSD in Poland among clinical and non-clinical samples.
- The two-factor second order model was supported as the best fit to the data.

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síntomas bajos con puntuaciones bajas en todos los síntomas.

Conclusiones: La versión polaca corregida del ITQ tiene una buena consistencia interna y propiedades psicométricas. El modelo se adapta bien para diferenciar los diagnósticos de TEPT/TEPTc según la CIE-11 y es una medida fiable de apoyo para hacer los diagnósticos de TEPT/TEPTc en la realización de una entrevista clínica.

And so the Steppenwolf had two natures, a human and a wolfish one. This was his fate, and it may well be that it was not a very exceptional one.

Herman Hesse, *Steppenwolf* (2004, p. 32)

1. Introduction

The eleventh revision of the International Classification of Diseases and Related Health Problems (ICD-11) has introduced the new diagnosis, complex PTSD (6B41, cPTSD). The World Health Organization distinguished complex PTSD from PTSD (6B40) (ICD-11 for Mortality and Morbidity Statistics). The term ‘complex PTSD’ was originally proposed by Judith Herman in 1992, as the result of her studies on the difficulties experienced by survivors of trauma in the following contexts: relational trauma in childhood, wartime experiences and domestic violence in adulthood (Herman, 2020). The ICD-11 PTSD and cPTSD diagnoses share the gate criterion of exposure to traumatic event(s) (ICD-11 for Mortality and Morbidity Statistics). PTSD is defined by re-experiencing (Re) such things as intrusive nightmares and flashbacks of the event(s), avoidance (Av) of internal and external stimuli associated with the trauma, and a sense of current threat (Th). These symptom clusters are jointly referred to as Post-Traumatic Stress Disorder (PTSD). Alongside these symptoms, cPTSD is defined by symptoms of affect dysregulation (AD), such as difficulties with calming down and returning to normal functioning, negative self-concept (NSC) with negative beliefs about oneself as worthless, accompanied by a sense of shame, guilt or failure in connection with the traumatic event, and disturbances in relationships (DR), persistent difficulties in maintaining relationships and building closeness with others. These last three symptom clusters are jointly referred to as Disturbances in Self-Organization (DSO) (Hyland, Shevlin, Elklit, et al., 2017; Shevlin et al., 2017). One notable difference between earlier formulations of complex PTSD and ICD-11 cPTSD relates to dissociation. In pre-ICD-11 descriptions of complex PTSD, dissociation was identified as a core feature of the condition (Herman, 2020). In ICD-11, dissociative symptoms are part of the overall symptom profile (i.e. dissociative flashbacks within the reexperiencing cluster and emotional numbing within the affective dysregulation cluster), but there

is no distinct dissociation symptom cluster. Thus, dissociation is not part of the essential requirements for a diagnosis of ICD-11 cPTSD (Hyland et al., 2024). However, research indicates (Hyland et al., 2024) that people meeting the diagnostic criteria for ICD-11 cPTSD experience higher levels of dissociation than those meeting the diagnostic criteria for ICD-11 PTSD or no diagnosis (Hyland et al., 2024).

To facilitate the differential diagnosis of PTSD and cPTSD, Cloitre, Shevlin, et al. (2018) developed The International Trauma Questionnaire (ITQ). This self-report measure is designed to reflect two overarching constructs, PTSD and DSO, with items nested in six subordinate symptom clusters. The clinical usefulness and good psychometric properties of the ITQ have been confirmed in worldwide research in clinical (Cloitre, Shevlin, et al., 2018; Karatzias et al., 2016; Sele et al., 2020), and non-clinical (Ben-Ezra et al., 2018; Cloitre et al., 2021; Owczarek et al., 2020; Rossi et al., 2022) samples. A substantial number of studies have identified the two-factor second-order model (PTSD and DSO, each measured by three first-order factors) as the best fit to the data (Hyland, Shevlin, Elklit, et al., 2017; Shevlin et al., 2017). A systematic literature review (Redican et al., 2021), including thirty-three studies of the ITQ in Europe, Africa, Asia and the United States shows that the ITQ differentiates the symptoms of ICD-11 PTSD and cPTSD.

The aim of the study was to adapt the ITQ for use in Polish language, assess the construct validity of ICD-11 PTSD and cPTSD using the amended version of Polish ITQ, and examine the convergent and discriminant validity of the Polish ITQ in non-clinical and clinical samples. In the course of the study, we developed the amended Polish version of ITQ that was translated according to the WHO guidelines (WHODAS 2.0 Translation package). Supplementary Table 1 lists the original English items and their adaptations into Polish.

2. Materials and methods

2.1. Procedure

The study was approved by the Bioethics Committee of the Institute of Psychiatry and Neurology in Warsaw, Poland (Resolution No. 35/2022). It was conducted from January to December 2023, in accordance with the Declaration of Helsinki (World

Medical Association, 2013), and national regulations. Information about the study was disseminated among adult psychiatric patients at the Institute of Psychiatry and Neurology and the Nowowiejski Hospital (clinical sample), and on social media, including profiles of the main researcher and the Akceptacja4FM.Lab project to Polish internet users aged 18–60 (non-clinical sample). Participants representing the clinical sample were native Polish adults aged 18–60, psychiatric patients in a stable mental state as assessed by a psychiatrist, enabling them to sign informed consent and fill in the questionnaires. Participants from non-clinical sample were also native Polish adults aged 18–60, who were not hospitalised for psychiatric reasons. For both samples, the information about the study included a description of its aims, procedure, voluntary participation, and the right to withdraw consent to participate in the study at any time. Participants were also informed about confidentiality, possible mood alteration when answering questions, and places where they could get help. Both groups were provided with informed consent forms, in online (non-clinical sample) and paper-and-pencil (clinical sample) formats. Participants from the clinical group were informed at the recruitment stage about their right to refuse to participate in the study, as well as the right to withdraw from it at any stage, and that a decision to not participate or to withdraw would not affect the further course of their treatment. Both groups completed the same set of questionnaires, in the different formats described. The respondents did not receive any gratification.

3. Instruments

3.1. The International Trauma Questionnaire (ITQ)

The ITQ, developed by Cloitre, Shevlin, et al. (2018), translated into Polish by Draczyńska, Nowakowska, Anczewska, is a self-report measure for the assessment of PTSD/cPTSD. The amended Polish version used for this study was translated in accordance with the WHO guidelines (WHODAS 2.0 Translation package). The questionnaire was translated by a team of native Polish speakers (experienced in psychiatric diagnostics) and backtranslated into English in a blind, written form to ensure accuracy. It consists of 19 items: the first, an open question, is about the necessary criterion of a history of traumatic experience, six pertain to PTSD (P1-P6), six to DSO (C1-C6), and there are six additional symptom-related functional impairment questions for PTSD (P7-P9) and DSO (C7-C9). Apart from the first question, the respondents answer on a five-point Likert scale, from 0 = 'Not at all' to 4 = 'Extremely'. A score of ≥ 2 is a confirmation of

a symptom or functional impairment. PTSD requires that at least one symptom is present from each of three symptom clusters: (1) re-experiencing the traumatic event in the present, (2) avoidance, (3) current sense of threat, and at least one symptom as part of functional impairment associated with these symptoms. cPTSD requires the confirmation of PTSD and at least one of two symptoms in each of the three symptom clusters for Disturbances in Self-Organization (DSO): (1) affect dysregulation, (2) negative self-concept, (3) disturbances in relationships. If the person examined meets the criteria for cPTSD, a diagnosis of PTSD is excluded.

4. Reference measures

4.1. The Traumatic Experience Checklist

The Traumatic Experience Checklist (TEC), (Nijenhuis et al., 2002), translated into Polish by Pietkiewicz and Tomalski (Nijenhuis et al., 2002), Cronbach's $\alpha = 0.72$. It contains 29 questions about various types of traumatic experiences, including events that are in line with the definition of traumatic event in the ICD-11:

exposure to an event or series of events of an extremely threatening or horrific nature, most commonly prolonged or repetitive events from which escape is difficult or impossible (e.g. torture, slavery, genocide campaigns, prolonged domestic violence, repeated childhood sexual or physical abuse). (ICD-11 for Mortality and Morbidity Statistics)

It also includes questions about potentially traumatic experiences: parentification, emotional neglect and abuse, bizarre punishments, and other stressful events. The tool indicates the context of the place of traumatic experiences: family of origin (parents, siblings), extended family (uncles, aunts, cousins, grandparents) and people outside the family (neighbours, friends, adoptive parents, teachers) (Nijenhuis et al., 2002). Respondents answer 'Yes' or 'No' to the question about a specific traumatic event, and specify the impact of this event on them (from 1 – 'none' to 5 – 'an extreme amount'), the age at which the event occurred and the level of support obtained (from 0 – 'none' to 2 – 'good').

4.2. The PTSD Checklist for DSM-5

To check convergent validity and correlation with PTSD according to DSM-5, the PCL-5 questionnaire (PTSD Checklist for DSM-5) was used, as translated by Zawadzki, Popiel, Białecka and Prąglowska (PCL-5, 2013). Cronbach's $\alpha = 0.96$. The questionnaire measures the severity of PTSD symptoms in adults according to the diagnostic criteria of DSM-5, namely re-experiencing, avoidance, negative cognitions and

mood, and arousal. The subjects indicate the severity of their symptoms on a 5-point scale.

4.3. The PSWQ Questionnaire

Correlation with anxiety was measured using the Penn State Worry Questionnaire (PSWQ), adapted by Janowski, which consists of 16 items, Cronbach's $\alpha = 0.94$. Respondents assess on a 5-point scale to what extent the behaviour, described by a given statement, is typical for them. Worry has been defined as a series of uncontrollable thoughts and images that lead to the formation of negative emotions and affect the development and maintenance of a permanent level of anxiety (Solarz & Janowski, 2013).

4.4. The CESD-R Scale

The CESD-R Scale (Center for Epidemiologic Studies Depression Scale – Revised), adapted by Koziara, consisting of 20 items, Cronbach's $\alpha = 0.95$, was used to measure depression (Koziara, 2016). For each statement describing mood or behaviour, subjects choose one of five answers regarding the frequency of their occurrence. The severity of depressive symptoms is determined based on the result.

5. Study participants

The sets of questionnaires were collected from $N = 452$ respondents aged 18–60; $N = 314$ from the non-clinical sample and $N = 138$ from the clinical sample. $N = 111$ of the respondents were psychiatrically hospitalised patients in a stable mental state at the Institute of Psychiatry, and $N = 27$ from the Nowowiejski Hospital. Following this, additional verification regarding the exclusion criteria and completion of data was performed. None of the records were discarded from the non-clinical sample and all $N = 314$ were included in the final analysis. For the clinical sample, eight respondents did not answer all of the ITQ questions, three did not confirm any potentially traumatic events in the past, so $N = 127$ respondents from the clinical sample were included in the analysis. Comparison of the sociodemographic data of the non-clinical and the clinical sample is presented in Table 1.

A total of 452 subjects participated in the study – 354 female, 76 males and 11 with other sex, 441 of whom provided complete data. In both samples, most participants were female: 88% in the non-clinical sample and 67% in the clinical sample. Several participants defined their sex as 'Other': 2% in the non-clinical and 3% the clinical sample. Mean age was 31.21 (non-clinical sample) and 35.17 (clinical sample). Most of the participants were from a city with over 500,000 inhabitants: 31.8% in the non-clinical and 70% in the clinical sample. The high percentage in

the non-clinical group is related to the location of the psychiatric clinic in the hospital where the study was conducted. In the non-clinical sample 55% individuals were married or in an informal relationship and 48% had a degree; in the clinical sample 56% of the participants were single, and 56% had a secondary level education. All diagnoses in the clinical sample were official primary ones, collected from the medical documentation of the hospitalised individuals who consented to participate in the study.

6. Statistical analysis

The statistical analysis was conducted in IBM SPSS Statistics 29, IBM SPSS Amos 26, and Microsoft Excel version 16.88. Each binomial variable was presented as prevalence in the sample, as both absolute (number of observations, N) and relative scores (percentage, %). Each continuous variable was characterised as a mean with standard deviation. The normality of the distribution was verified with the Shapiro–Wilk W test and by analysis of histograms, skewness, and kurtosis. The heterogeneity of variance between the subgroups was checked with Levene's test. Intergroup comparisons were conducted through the analysis of variance or with Welch's t -test; the assumption of homogeneity of variance was not met. For the contingency tables a chi-squared test was utilised. Fisher's exact test was performed in cases of an expected count of less than five in at least one of the cells.

A series of confirmatory factor analyses (CFAs) was performed to evaluate the construct validity of the ITQ, which is an approach replicating previous similar studies (Andisha et al., 2023; Ho et al., 2019; Kazlauskas et al., 2020). Seven models were tested. Model 1 is a single-factor model with all symptoms loading on a single latent factor (cPTSD). Model 2 is a correlated six factor first-order model. Model 3 includes six ITQ symptoms as first-order factors and a single second-order factor (cPTSD). Model 4 has two second-order factors, PTSD and DSO, each containing three respective symptoms as first-order factors. Model 5 is two second-order factors, in which DSO is divided into three first-order factors (AD, NSC, DR), and the PTSD latent factor has its respective items loaded directly. Model 6, has two second-order factors, in which PTSD is divided into three first-order factors (Re, Av, Th), and the DSO latent factor has its respective items loaded directly. Model 7 consists of two first-order factors, PTSD and DSO, each loaded directly with its respective items (Figure 1).

Each CFA model was verified regarding its goodness-of-fit with the calculations based on Kline's suggestions (Kline, 2011) for the determination of good model fit, which were followed for the CFA analyses: a chi-square-degrees of freedom (χ^2/df) ratio of less

Table 1. Comparison of the sociodemographic data of the non-clinical and the clinical sample.

		Non-clinical (N = 314)	Clinical (N = 127)	
Age, M ± SD		31.21 ± 10.83	35.17 ± 12.72	$F = 10.897, p = .001$
Sex, N (%)	Female	269 (86%)	85 (67%)	$\chi^2 = 19.719, p < .001^*$
	Male	38 (12%)	38 (30%)	
	Other	7 (2%)	4 (3%)	
Marital status, N (%)	Single	114 (35%)	71 (56%)	$\chi^2 = 25.154, p < .001^*$
	Married	72 (23%)	28 (22%)	
	Informal relationship	100 (32%)	17 (13%)	
	Divorced	26 (8%)	7 (6%)	
	Separated	1 (1%)	1 (1%)	
	Widowed	1 (1%)	3 (2%)	
Level of education, N (%)	Primary	10 (3%)	7 (6%)	$\chi^2 = 7.317, p = .054^*$
	Secondary	148 (47%)	71 (56%)	
	Vocational	5 (2%)	4 (3%)	
	Higher	151 (48%)	45 (35%)	
Place of residence, N (%)	Rural area	48 (15%)	9 (7%)	$\chi^2 = 42.000, p < .001$
	City <50k	41 (13%)	16 (13%)	
	City 50–150k	42 (13%)	4 (3%)	
	City 150–500k	65 (21%)	10 (8%)	
	City >500k	118 (38%)	88 (70%)	
With whom live, N (%)	Alone	42 (14%)	33 (27%)	$\chi^2 = 11.026, p = .004$
	With family	180 (57%)	60 (48%)	
	With other persons	92 (29%)	31 (25%)	
Source of income, N (%)	Full-time job	127 (40%)	41 (32%)	$\chi^2 = 43.434, p < .001^*$
	Mandate (civil-law) contract	51 (16%)	13 (10%)	
	Sickness pension	8 (3%)	13 (10%)	
	Retirement pension	0 (0%)	4 (3%)	
	Social services	4 (1%)	12 (9%)	
	Family	69 (22%)	20 (16%)	
	Other	55 (18%)	24 (19%)	
Primary diagnosis, N (%)	F00-09		2 (2%)	
	F10-19		4 (3%)	
	F20-29		12 (9%)	
	F30-39		39 (30%)	
	F40-49		24 (19%)	
	F50		1 (1%)	
	F60-61		43 (34%)	
	F80-89		2 (2%)	

Note: N: number of observations; M: mean; SD: standard deviation; F: Fisher's Snedecor test results from one-way analysis of variance; p: probability in the test; χ^2 : chi-squared test result or *Fisher's exact test result due to expected count of 5 or less in at least one cell; F00-09: Organic mental disorders; F10-19: Substance use disorders; F20-29: Schizophrenia and related disorders; F30-39: Mood disorders; F40-49: Neurotic, stress-related and somatoform; F50: Eating disorders; F60-61: Personality disorders; F80-89: Disorders of psychological development.

than 3:1 indicates good model fit; Comparative Fit Index (CFI) and Tucker Lewis Index (TLI) values greater than 0.90 reflect acceptable model fit, and values greater than 0.95 reflect excellent fit; a root-mean-square error of approximation (RMSEA) values 0.08 or less and 0.05 or less indicate an acceptable and excellent model fit respectively; and RMSEA with 90% confidence intervals (RMSEA 90% CI) values of 0.05 or less reflects an excellent model fit, while values less than 0.10 reflect an acceptable fit (Hansen et al., 2015).

Latent Class Analysis (LCA) was performed based on six binary variables (Av, Re, Th, AD, NSC, DR), defined as clinically significant symptoms of cPTSD (i.e. a score of at least two in at least one of two items in a factor). Six consecutive models with the number of classes from two to six. The best-fit model was chosen based on the results of the use of the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and the chi-squared test. Lower AIC and BIC values indicate better model fit. The model with the lowest AIC and BIC was considered to be the best model (Akaike, 1987; Raftery, 1995).

Nine general linear models (GLMs), with each of the ITQ dimensions, PTSD score, DSO score, and

total ITQ score as dependent variables, were constructed. The predictors comprised the PTSD symptoms assessed with the PCL-5 scales, PSWQ total score, with the depressive symptoms assessed with CESD-R. Each model was comprised of the sex, age and group of the participants as possible confounding factors. The goodness-of-fit of the GLMs was verified with a global F test. Analysis of residuals was performed to assess the validity of assumptions of normality, and homoscedasticity (with the modified Breusch–Pagan Test).

Effect sizes, in statistical terms, were assessed in several ways: for GLMs (coefficient of determination R^2), for each parameter in the GLM (standardised β parameter), and for each variable as factor loading in CFA. Those quotients may be interpreted in terms of Cohen's thresholds for weak (0.1), moderate (0.3) and strong effect sizes (0.5). The level of significance was adopted for $\alpha = 0.05$.

Factor models of the post-traumatic stress disorder (PTSD) and complex PTSD structure, tested in confirmatory factor analysis based on the International Trauma Questionnaire scores in the studied group in both samples, are presented in Figure 1.

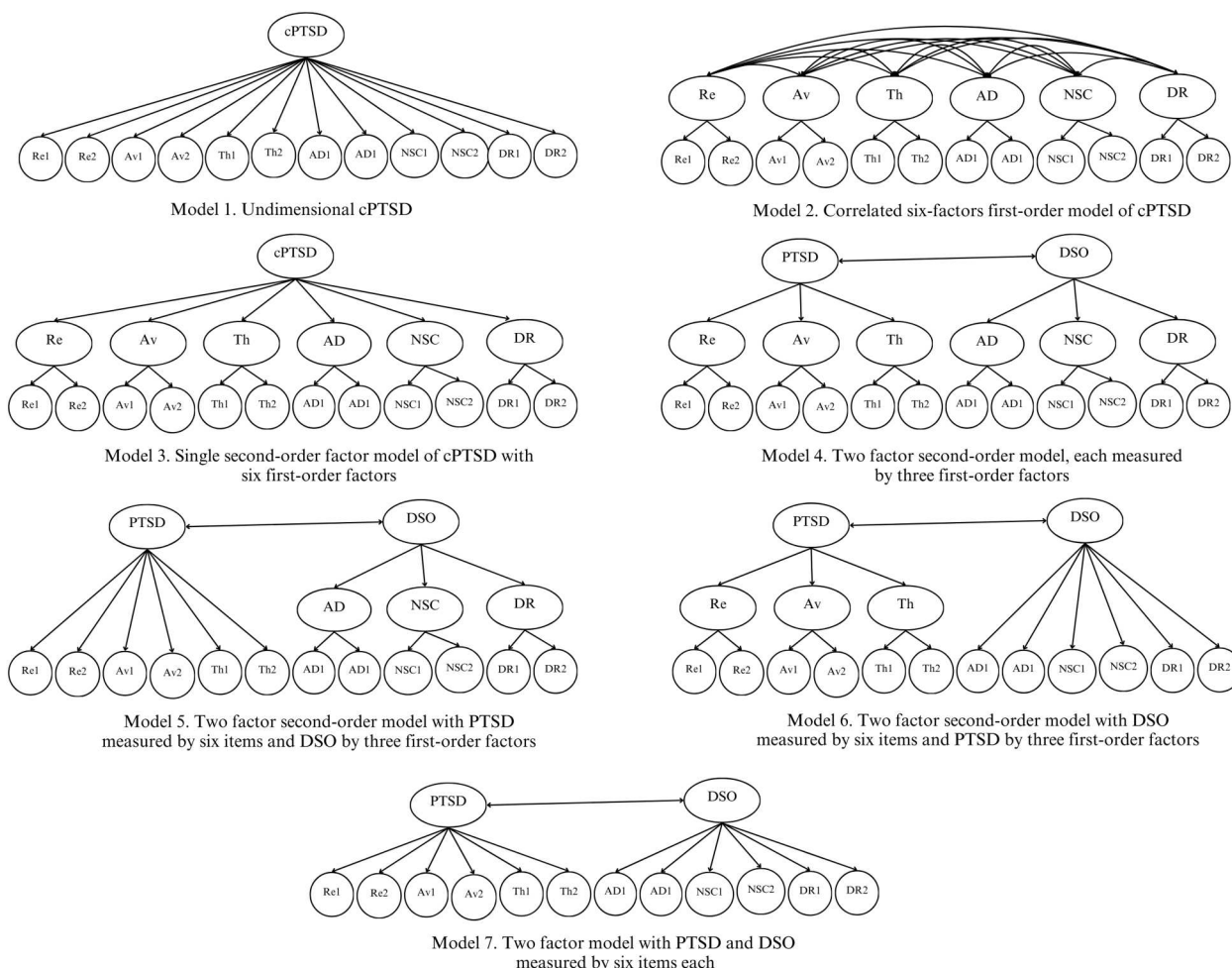


Figure 1. Factor models of the Post-traumatic Stress Disorder (PTSD) and complex PTSD structure, tested in confirmatory factor analysis based on the International Trauma Questionnaire scores in the studied group with non-clinical and clinical samples.

7. Exposure to potentially traumatic events

The mean of potentially traumatic lifetime events was greater in the clinical sample than in the non-clinical sample in the TEC questionnaire, but both numbers exceeded nine ($M = 10.20 \pm 4.76$ vs $M = 9.04 \pm 4.95$, respectively; $F = 4.578$, $p = .033$). There were significant differences in the frequencies of cases reporting being a second-generation war victim, emotional neglect by more distant family members, emotional abuse by non-family members, and sexual abuse by non-family members. Detailed results are presented in Table 2.

The three experiences (bold in Table 2) with the highest rate of representation in both groups were (non-clinical and clinical sample): family problems 67% and 76%, emotional neglect 80% and 83% and emotional abuse 70% and 74% both by your parents, brothers or sisters, respectively.

8. Results

8.1. Intergroup comparisons

The total prevalence of PTSD and cPTSD was 7.8% ($N = 34$) and 38.9% ($N = 170$) respectively, according to

the ITQ scoring criteria in the group as a whole. In the non-clinical sample those frequencies were PTSD 7.3% ($N = 23$) and cPTSD 36.3% ($N = 114$), while in the clinical sample PTSD 8.9% ($N = 11$) and cPTSD 45.5% ($N = 56$). The differences were non-significant ($\chi^2 = 4.178$, $df = 2$, $p = .124$).

In the non-clinical sample, the mean ITQ total score, PTSD score, DSO score, Re, Av, AD, NSC, and DR scores were significantly higher in the clinical group than in the non-clinical sample. The difference in the sense of threat was nonsignificant. Detailed results of the intergroup comparisons can be found in Table 3.

8.2. Confirmatory factor analysis results – construct validity

Models 1 (unidimensional cPTSD), 6 (two factor second order with DSO measured by six items and PTSD by three first-order factors) and 7 (two factor with PTSD and DSO measured by six items each) were rejected due to poor fit indices in both samples. Model 3 (single second-order with six first-order factors) and 5 (two factor second-order with PTSD

Table 2. Comparison of the frequencies of potentially traumatic life events reported in the Trauma Evaluation Checklist by the non-clinical and the clinical sample in the studied group of individuals exposed to potentially traumatic life events.

	Non-clinical (N = 314)		Clinical (N = 127)		χ^2	p
	N	%	N	%		
1. Having to look after your parents and/or brothers and sisters when you were a child.	136	57%	65	52%	2.719	0.112
2. Family problems (e.g. parent with alcohol or psychiatric problems, poverty).	216	67%	97	76%	2.527	0.132
3. Loss of a family member (brother, sister, parent) when you were a CHILD.	68	22%	38	30%	3.555*	0.065
4. Loss of a family member (child or partner) when you were an ADULT.	89	28%	43	34%	1.432	0.251
5. Serious bodily injury (e.g. loss of a limb, mutilation, burns).	55	18%	31	24%	2.737	0.111
6. Threat to life from illness, an operation, or an accident.	81	26%	32	25%	0.017	0.905
7. Divorce of your parents.	83	26%	43	34%	2.443	0.131
8. Your own divorce.	39	12%	12	10%	0.693	0.418
9. Threat to life from another person (e.g. during a crime).	63	20%	26	21%	0.018	0.896
10. Intense pain (e.g. from an injury or surgery).	134	43%	54	43%	0.001	0.999
11. War-time experiences (e.g. imprisonment, loss of relatives, deprivation, injury).	8	3%	6	5%	1.394*	0.241
12. Second generation war – victim (war-time experiences of parents or close relatives).	26	8%	19	15%	4.527	0.038
13. Witnessing others undergo trauma.	169	54%	79	63%	2.881	0.111
14. Emotional neglect (e.g. being left alone, insufficient affection) by your parents, brothers or sisters.	251	80%	105	83%	0.437	0.594
15. Emotional neglect by more distant members of your family (e.g. uncles, aunts, nephews, nieces, grandparents).	116	37%	62	49%	5.298	0.024
16. Emotional neglect by non-family members (e.g. neighbours, friends, stepparents, teachers).	115	37%	58	46%	3.335	0.084
17. Emotional abuse (e.g. being belittled, teased, called names, threatened verbally, or unjustly punished) by your parents, brothers or sisters.	221	70%	94	74%	0.585	0.486
18. Emotional abuse by more distant members of your family.	149	48%	48	38%	3.412	0.072
19. Emotional abuse by non-family members.	135	43%	75	59%	9.352	0.003
20. Physical abuse (e.g. being hit, tortured, or wounded) by your parents, brothers, or sisters.	165	53%	70	56%	0.327	0.598
21. Physical abuse by more distant members of your family.	28	9%	14	11%	0.539	0.475
22. Physical abuse by non-family members.	96	31%	41	32%	0.124	0.734
23. Bizarre punishment.	88	28%	46	38%	3.867	0.051
24. Sexual harassment (acts of a sexual nature that DO NOT involve physical contact) by your parents, brothers, or sisters.	41	13%	16	13%	0.01	0.999
25. Sexual harassment by more distant members of your family.	45	14%	9	7%	4.315	0.053
26. Sexual harassment by non-family members.	109	35%	43	34%	0.014	0.912
27. Sexual abuse (unwanted sexual acts involving physical contact) by your parents, brothers, or sisters.	19	6%	9	7%	0.163	0.83
28. Sexual abuse by more distant members of your family.	20	6%	4	3%	1.822	0.247
29. Sexual abuse by non-family members.	74	24%	44	35%	6.156	0.017

Note: N: number of observations; χ^2 : chi-squared test result or *Fisher's exact test result due to expected count of 5 or less in at least one cell, p: probability in the test.

measured by six items and DSO by three first-order factors) showed acceptable CFI and TLI in the non-clinical sample, but not in the clinical sample and thus were rejected. Model 4 (two factor second-order each measured by three first order factors) showed excellent TFI and CFI and an acceptable RMSEA in the non-clinical sample, but those indices were poor-to-acceptable in the case of the clinical sample. Model 4 (two factor second-order each measured by three first order factors) presented excellent CFI and TLI, and the 90%CI of RMSEA was within the acceptable range. Based on the theoretical assumptions and the highest fit parameters (consistently in both study samples), Model 4 (two factor second-order each measured by three first order factors), which is consistent with the ICD-11 proposals for PTSD and cPTSD, was chosen as the best-fitting one. However, it should be noted that the indices of model 2 (correlated six-factors first order) and Model 4 (two factor second-order each measured by three first order factors) in the non-clinical sample are very similar. In model 4 (two factor second-order), all factor loadings were statistically significant and reached a satisfactory level (at least 0.5). The lowest loading was seen in the

case of item AD1, which can be contributed to its very general content (Table 4). In the case of the clinical samples, items Re1 and Th1 had lower and items Re2 and DR1 had higher factor loadings than in the non-clinical sample. The remaining factor loadings were similar between the groups (Table 5).

Factor loadings of Model 4 are shown in Table 4. The factor loading in both samples were mostly high (>0.6), with an exception for Re1 in the clinical sample and AD1 in both the non-clinical and clinical samples, which were acceptable (>0.4).

In the current study, the internal consistency indicators for PTSD ($\alpha = 0.810$), DSO ($\alpha = 0.875$) and total ITQ scale ($\alpha = 0.887$) were good.

8.3. Latent class analysis results – construct validity

Latent class analysis (LCA) was used to determine the optimal number of classes based on responses to the ICD-11 cPTSD. LCA was performed to determine the appropriate number of classes based on the probability of meeting the diagnostic thresholds for the three PTSD symptom clusters (Re, Av, and Th) and

Table 3. Comparison of the International Trauma Questionnaire, PTSD checklist, Penn State Worry Questionnaire and Center for Epidemiologic Studies Depression scale – revised total scores and subscores between the non-clinical and clinical samples in the studied group of persons exposed to potentially traumatic life events.

		Non-clinical (N = 314)		Clinical (N = 127)		F	p
		M	SD	M	SD		
International Trauma Questionnaire	Re-experiencing	3.04	2.41	4.17	2.38	19.926	<.001
	Avoidance	4.07	2.68	4.69	2.53	4.994	.026
	Sense of threat	4.22	2.54	4.62	2.42	2.296	.130
	Affective dysregulation	4.51	2.15	5.22	2.13	9.813	<.001
	Negative self-concept	3.87	2.77	5.37	2.57	7.392*	<.001
	Disturbances in relationships	4.39	2.66	5.14	2.59	7.239	.007
	PTSD score	11.33	6.31	13.48	5.36	13.043*	<.001
PTSD Checklist	DSO score	12.78	6.61	15.73	6.15	18.774	<.001
	total score	24.11	11.53	29.21	10.08	21.255*	<.001
	Intrusion	8.05	5.63	11.04	5.29	26.398	<.001
	Avoidance	9.94	5.59	12.49	5.09	21.400*	<.001
	Negative changes in cognition and emotions	9.96	5.74	12.88	4.93	28.818*	<.001
Penn State Worry Questionnaire score	Changes in arousal and activity	9.52	5.32	12.09	4.63	25.431*	<.001
	total score	37.47	20.03	48.50	17.11	33.925*	<.001
		56.92	11.57	58.48	14.02	1.450	.229
Center for Epidemiologic Studies Depression scale – revised	Sadness	6.16	3.78	8.10	3.58	24.514	<.001
	Anhedonia	3.20	2.75	4.82	2.75	31.348	<.001
	Appetite	1.59	2.22	2.35	2.59	8.497*	.004
	Sleep	4.82	3.47	5.80	3.40	7.408	.007
	Thinking and concentration	4.05	2.69	5.43	2.69	23.024	<.001
	Guilt	3.29	2.86	4.76	3.01	23.024	<.001
	Tired	4.10	2.51	5.21	2.55	17.620	<.001
	Movement	2.26	2.22	3.40	2.26	23.585	<.001
	Suicidal ideation	1.80	2.48	2.97	0.26	20.196*	<.001
total score		31.27	19.63	43.02	18.91	33.076	<.001

Note: PTSD: post-traumatic stress disorder; DSO: Disorganized self-organisation; M: mean value; SD: standard deviation; N: number of observations; F: F test statistics in the analysis of variance; p: probability in the test.

Table 4. Parameter of the goodness-of-fit of the confirmatory factor analysis models of the International Trauma Questionnaire structure, dividend into the non-clinical and clinical samples in the studied group of individuals exposed to potentially traumatic life events.

		TLI	CFI	RMSEA	RMSEA 90% CI		χ^2 (df) p
Non-clinical sample (N = 314)	Model 1	0.697	0.752	0.163	0.150	0.176	501.801 (54) p < .001
	Model 2	0.965	0.979	0.056	0.037	0.074	76.670 (39) p < .001
	Model 3	0.907	0.933	0.090	0.076	0.105	170.072 (48) p < .001
	Model 4	0.965	0.975	0.055	0.038	0.072	91.655 (47) p < .001
	Model 5	0.913	0.934	0.087	0.073	0.102	169.710 (50) p < .001
	Model 6	0.886	0.914	0.100	0.086	0.114	206.174 (50) p < .001
	Model 7	0.841	0.872	0.118	0.105	0.132	283.701 (53) p < .001
		TLI	CFI	RMSEA	RMSEA 90% CI		χ^2 (df) p
Clinical sample (N = 127)	Model 1	0.568	0.646	0.180	0.160	0.202	275.488 (54) p < .001
	Model 2	0.928	0.958	0.074	0.041	0.104	65.549 (39) p < .001
	Model 3	0.886	0.917	0.093	0.067	0.118	100.098 (48) p < .001
	Model 4	0.893	0.924	0.090	0.063	0.116	94.652 (47) p < .001
	Model 5	0.788	0.839	0.126	0.104	0.150	150.718 (50) p < .001
	Model 6	0.737	0.800	0.141	0.118	0.164	175.011 (50) p < .001
	Model 7	0.646	0.715	0.163	0.142	0.185	231.322 (53) p < .001

Note: N: number of observations; TLI: Tucker–Lewis index; CFI: comparative fit index; RMSEA: root mean square error of approximation; CI: confidence interval; χ^2 : chi-squared test; df: degrees of freedom; p: probability in the test.

Table 5. Factor loadings of the items of the International Trauma Questionnaire in relationship to the factors in the first-order six factor model (Model 4: two factor second-order each measured by three first order factors), divided into the non-clinical and clinical samples in the studied group of patients exposed to potentially traumatic life events.

Item	Factor	Non-clinical (N = 314)	Clinical (N = 127)
Re1. Upsetting dreams reminding of the harm that happened.	Re	0.733	0.553
Re2. Images and memories with the feeling they are happening right now.		0.808	0.979
Av1. Avoiding internal cues (e.g. thoughts) about the trauma.	Av	0.725	0.720
Av2. Avoiding external reminders of what happened.		0.827	0.775
Th1. Being on alert, watchful, on guard.	Th	0.789	0.647
Th2. Excessive startle response.		0.745	0.834
AD1. Having problems with calming down when upset.	AD	0.509	0.514
AD2. Feeling unable to have emotions or empty.		0.790	0.735
NSC1. Feeling like a failure.	NSC	0.913	0.952
NSC2. Feeling worthless.		0.871	0.807
DR1. Feeling distant or cut off from others.	DR	0.849	0.989
DR2. Having difficulties staying close to others.		0.822	0.784

Note: All shown factor loadings are significant ($p < .001$). N: number of observations; Re: Re-experiencing; Av: Avoidance; Th: Sense of threat; AD: Affective dysregulation; NSC: Negative self-concept; DR: Disturbances in relationships.

the four DSO symptom clusters (Hy, Ho, NSC, and DR). Six latent class models were assessed (one through six classes) to determine optimal fit. Models with 1–6 latent classes were estimated using the robust maximum likelihood estimation (Yuan & Bentler, 2000). The relative fit of these models was assessed using the Akaike Information Criterion (1987), the Bayesian Information Criterion (Raftery, 1995); the BIC is considered the best test for detecting the correct number of classes. In each case, the model with the lowest value is considered the best fitting model. When a nonsignificant value ($P > .05$) occurs, this suggests that the model with one less class should be accepted. Higher entropy values indicated greater classification accuracy. The LCA model fit statistics are presented in Table 5. Log-likelihood was the highest for the 5-classes model. The AIC value supported a 4-classes model. The AIC and BIC values supported a 4-classes model. The AIC was marginally lower for the 5-classes model than for 4-classes model, but the chi-squared test result was statistically insignificant, meaning that one of the neighbouring models should be taken into consideration. The BIC was lower for the 3-classes model than for 4-classes model. All in all, based on the previous findings, the theoretical assumptions, and only slight differences in case of fit indices with the 3 and 5 – classes models, it was decided that the 4-classes model was best fitted to the empirical data. Entropy of the 4-classes model was 0.819, which indicated a high separation of the classes. Goodness-of-fit indices in the latent class analyses of the six-factor first-order model of cPTSD,

based on the International Trauma Questionnaire scores in the non-clinical and clinical samples, are presented in Table 6.

Figure 2 shows a graphical presentation of the latent classes of the chosen model. Class one was considered baseline as it has the lowest probabilities of meeting all six diagnostic criteria for cPTSD based on the ITQ. 5% of the research sample belong to this class. The largest class (67%) was defined as cPTSD, with the highest probability of fulfilling all six criteria. The next class (17%) had a high probability of meeting the three PTSD criteria, with a relatively high probability of fulfilling the affective dysregulation criteria and lower probability of fulfilling the negative self-concept and disturbances in relationships criteria – the class was thus defined as PTSD. The last class (11%) had relatively high probability of fulfilling three DSO criteria, but a low one when it came to PTSD criteria, and thus was defined as a DSO class. The probability of belonging to classes cPTSD, PTSD, and DSO compared to the baseline class were higher in the clinical sample than in the non-clinical sample (all three comparisons $p < .001$).

8.4. General linear models – convergent and discriminant validity

All of the general linear models constructed, predicting the scores of the ITQ dimensions, fit the empirical data. The variance of the errors did not depend on the values of the independent variables ($p > 0.05$). All models seemed to be eligible for the purposes of

Table 6. Goodness-of-fit indices in the latent class analyses of the six-factor first-order model of complex post-traumatic syndrome disorder based on the International Trauma Questionnaire scores in the non-clinical and clinical samples.

	Log-likelihood	AIC	BIC	df	χ^2	$\chi^2 p$	G ²	G ² p	Entropy
2 classes	–1229	2486	2543	14	104,814	<.001	102.741	<.001	0.784
3 classes	–1211	2466	2556	22	61,545	<.001	67.337	<.001	0.809
4 classes	–1195	2450	2572	30	49,422	.014	34.669	.255	0.819
5 classes	–1187	2449	2605	38	28,402	.871	29.853	.871	0.863
6 classes	–1199	2489	2677	46	35,513	.868	17.640	.999	0.632

Note: AIC: Akaike Information Criterion; BIC: Bayesian Information Criterion; χ^2 : chi-squared test; df: degrees of freedom; p: probability in the test.

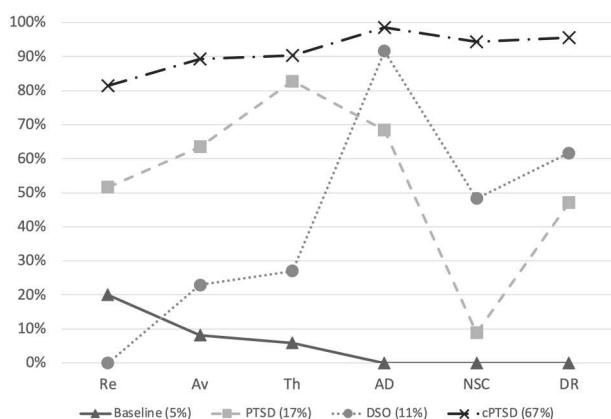


Figure 2. Latent classes of Post-traumatic Stress Disorder (PTSD), complex PTSD (cPTSD), and Disturbances in Self-Organization (DSO) based on International Trauma Questionnaire scores in the group comprising a non-clinical and clinical sample. Re: Re-experiencing; Av: Avoidance; Th: Sense of threat; AD: Affective dysregulation; NSC: Negative self-concept.

prediction of the value of the dependent variables (namely ITQ subscores), as the coefficients of determination (R^2) exceeded 0.4 (Table 7).

A rise in the ITQ Re-experiencing score was predicted by a rise in PCL-5 Intrusion score, with a large effect size. An increase in ITQ Avoidance was associated with an increase in PCL-5 Intrusion and Avoidance, both effects being of moderate size. A rise in a sense of threat was linked with a rise in PCL-5 Intrusion score (weak effect size), Changes in arousal and activity (strong effect size), and PSWQ score (very weak effect size). A rise in the Affective dysregulation score was predicted by an increase in PCL-5 Intrusion, Negative changes in cognition and emotions, Changes in arousal and activity, PSWQ score, and CESD-R Sadness score: the effects here

were weak. An increase in Negative self-concept score was associated with a rise in PCL-5 Negative changes in cognition and emotions (moderate effect size), PSWQ score (weak effect size), CESD-R Guilt (moderate effect size), and a fall in CESD-R Appetite score (very weak effect size). An increase in Disturbances in relationships score was predicted by the Negative changes in cognition and emotions score, with a strong effect size. A rise in the PTSD score in ITQ was associated with a rise in PCL-5 Intrusion (strong effect size), Avoidance (weak effect size), Changes in arousal and activity (weak effect size), and PSWQ score (weak effect size). An increase in the DSO ITQ score was linked to a rise in PCL-5 Avoidance (weak effect size), Negative changes in cognition and emotions (moderate effect), PSWQ score (weak effect), CESD-R Guilt (moderate effect), and a fall in the Appetite score (weak effect). ITQ total score was predicted by the scores in all PCL-5 scales (all of moderate effect size), PSWQ score (weak effect), CESD-R Guilt (moderate effect) and Appetite (weak effect) (Table 7).

9. Discussion

Our study is the first psychometric validation of the Polish version of ITQ questionnaire, carried out in a group comprising non-clinical and clinical samples, among medically stable psychiatric patients (according to the attending psychiatrist's opinion). It also presents data on the prevalence of ICD-11, PTSD and cPTSD in Poland. The total prevalence of ICD-11 PTSD was 7.8% and for cPTSD 38.9%, according to the ITQ scoring criteria in the studied group as a whole. In the non-clinical sample, the prevalence of PTSD was 7.3% and cPTSD 36.3%, while in the clinical

Table 7. Results of general linear models predicting the scores of the International Trauma Questionnaires scales, corresponding with the six symptoms of complex post-traumatic stress disorder (PTSD) in the group comprising non-clinical and clinical samples. Reported as standardised estimates (effect sizes).

		Re	Av	Th	AD	NSC	DR	PTSD	DSO	Total score
PCL-5	Intrusion	0.673 [†]	0.384 [†]	0.193 [†]	0.126*	-0.006	-0.021	0.515 [†]	0.030	0.295 [†]
	Avoidance	-0.109	0.362 [†]	0.124	0.128	0.086	0.105	0.165**	0.120*	0.158 [†]
	Negative changes in cognition and emotions	-0.028	-0.062	-0.129	0.233 [†]	0.310 [†]	0.533 [†]	-0.091	0.422 [†]	0.197 [†]
	Changes in arousal and activity	0.047	-0.044	0.689 [†]	0.160*	0.000	0.089	0.283 [†]	0.088	0.204 [†]
PSWQ score (Worry)		0.070	0.019	0.084*	0.125**	0.159 [†]	-0.074	0.085*	0.070*	0.080**
		0.001	-0.043	0.009	0.150*	0.067	-0.012	-0.015	0.073	0.034
CESD-R	Sadness	-0.025	-0.020	0.011	-0.086	0.078	0.083	-0.014	0.038	0.015
	Anhedonia	0.026	-0.053	-0.028	-0.061	-0.098**	-0.073	-0.024	-0.091**	-0.066**
	Appetite	0.088	0.033	-0.050	-0.050	0.058	0.039	0.029	0.024	0.030
	Sleep	-0.010	-0.061	-0.059	0.028	0.000	-0.017	-0.055	0.003	-0.028
	Thinking and concentration	0.028	0.001	-0.063	0.034	0.380 [†]	0.008	-0.015	0.176 [†]	0.094*
	Guilt	-0.012	0.119	-0.070	0.129	-0.053	0.118	0.018	0.067	0.049
	Tired	-0.036	0.067	0.011	-0.043	-0.058	-0.002	0.019	-0.040	-0.013
	Movement	0.087	-0.062	0.001	-0.047	-0.011	-0.055	0.007	-0.042	-0.020
	Suicidal ideation	24.590 [†]	16.178 [†]	27.436 [†]	57.932 [†]	54.108 [†]	23.834 [†]	47.979 [†]	65.194 [†]	96.261 [†]
	F	0.518	0.409	0.546	0.538	0.708	0.510	0.682	0.745	0.813
	R ²									

Note: All models are adjusted for the sex, age and group assignment (non-clinical vs clinical) of the participants. Re: Re-experiencing; Av: Avoidance; Th: Sense of threat; AD: Affective dysregulation; NSC: Negative self-concept; DR: Disturbances in relationships; PCL-5: PTSD Checklist; PSWQ: Penn State Worry Questionnaire; CESD-R: Center for Epidemiologic Studies Depression scale – Revised; F: the F test's statistics of overall fit of the model, R²: coefficient of determination, * $p < .05$, ** $p < .01$, [†] $p < .001$.

sample PTSD was at 8.9% and cPTSD 45.5%. The combined rate of ICD-11 PTSD and cPTSD in the study was 46.7%. The rates of PTSD and cPTSD differ between various studies (Ben-Ezra et al., 2018; Christen et al., 2021; Cloitre et al., 2013; Hyland, Shevlin, Elklit, et al., 2017; Sele et al., 2020; Skeie-Larsen et al., 2023). Among UK's male veterans with trauma experience (Hyland, Shevlin, Brewin, et al., 2017) the combined rate was 64.5%, with a PTSD rate of 10.9% and cPTSD rate of 53.6%. In a Norwegian clinical sample (Sele et al., 2020) of people with interpersonal trauma in childhood, the combined rate was 73.8% with PTSD at 13.4% and 60.4% diagnosed with cPTSD. Among individuals in the US seeking treatment at a trauma clinic for problems related to interpersonal violence (Cloitre et al., 2013), the combined rate was 71.9%, with 42.8% with cPTSD and 29.1% with PTSD. In the German general population, trauma-exposed adult sample (Christen et al., 2021) the combined diagnostic rate of PTSD (5.0%) and cPTSD (3.2%) was 8.2%, which is lower than in the mentioned studies but similar to trauma-exposed Israeli adults (Ben-Ezra et al., 2018), among whom was found a prevalence rate of PTSD of 9.0% and cPTSD of 2.6% with a combined prevalence rate of 11.6%. The most prevalent ICD-10 diagnosis in the clinical sample of our study was personality disorders (34%) and affective disorders (30%), which might explain the elevated proportion of cumulated PTSD and cPTSD. The results obtained are higher than in the study on the Polish population of Adult Children of Alcoholics (Rzesutek et al., 2024), where the results were 7.6% for ICD-11 PTSD and 6.9% for ICD-11 cPTSD, with a combined rate 14.5%. The selection of the target group can be a source of bias. People who have grown up in an alcoholic family are not necessarily ready to talk about their experiences. Rather, they focus on symptoms that cause the discrepancy between external and internal self-perception (Czabała, 2009). The prevalence of PTSD and cPTSD in different samples is subject to an ongoing scientific debate (Christen et al., 2021; Cloitre, Shevlin, et al., 2018). There is a hypothesis that non-clinical (community) rates of PTSD should be higher than cPTSD, while the reverse is true for trauma clinics (Brewin et al., 2009; Christen et al., 2021). However, this view is challenged by studies showing that multiple traumatic experiences are associated with higher rates of cPTSD than PTSD (Christen et al., 2021) and that in non-clinical samples multiple experiences of traumatisation are more common than a single experience (Christen et al., 2021). In our study the three most frequently indicated traumatic experiences were: emotional neglect 80% in the non-clinical and 83% in the clinical sample; emotional abuse at 70% the non-clinical and 74% in the clinical sample; family problems (e.g. parent with alcohol or psychiatric

problems, poverty) were at 67% in the non-clinical and 76% in the clinical sample. In a Norwegian clinical sample (Sele et al., 2020) exposed to interpersonal trauma in childhood, the most frequent experiences were of emotional abuse 86.1%, sexual abuse 77.2% and physical abuse 62.9%. In a British clinical sample of trauma-exposed patients (Hyland, Shevlin, Brewin, et al., 2017) the most frequently experienced trauma was physical assault 73.7%, followed by physical and/or sexual abuse during childhood 47.6%, and the most distressing traumatic events identified were childhood sexual abuse or molestation, at 15.5%.

Testing the construct validity of ICD-11 cPTSD in this study, both Model 2 (correlated six-factors first order) and Model 4 (two factor second-order each measured by three first order factors) fitted the data well in the non-clinical sample. Similarly, to other studies (Cloitre, Bisson, et al., 2018; Fresno et al., 2023; Haselgruber et al., 2020; Mordeno et al., 2019; Shevlin et al., 2018) that identified both models as being of equivocal fit or (Cloitre, Bisson, et al., 2018; Fresno et al., 2023) both models to be comparable in terms of model fit in non-clinical and clinical samples. It should be noted that clinical samples in other studies means samples of trauma-exposed patients in an outpatient setting: soldiers in the Armed Forces of the Philippines with at least two armed encounters against the rebel forces, mostly men (Mordeno et al., 2019), or male veterans mental health users from National Centre for Mental Health (Haselgruber et al., 2020). No previous study has researched a clinical group of psychiatrically hospitalised patients. Thus, the results of the current study may also be explained by the specificity of the clinical group investigated. Most studies (Ben-Ezra et al., 2018) indicate that the Model 4 (two factor second-order each measured by three first order factors) was the best-fit model. This model is also consistent with the ICD-11 conceptualisation of cPTSD. Based on the theoretical assumptions and the good fit indices for Model 4 in the non-clinical sample and good-to-acceptable fit of this model in the clinical sample, the second-order two-factor model correlated with six first-order factors model was assumed to be best fitted to the data.

Examining the psychometric properties, all factors in the first-order six factor model were statistically significant and high (>0.6) for the non-clinical and clinical samples with the exception of Re 'having upsetting dreams reminding of the harm that happened' in the clinical sample and AD-hyperactivation – 'having problems with calming down when upset' – in both the non-clinical and clinical samples, which were acceptable (>0.4). Other studies indicate similar results (Cyr et al., 2022; Fresno et al., 2023; Haselgruber et al., 2020). Next one (Fresno et al., 2023) reports in the range between $<0.6 > 0.4$ for AD-hyperactivation. Another (Haselgruber et al., 2020) reports that two

items exhibited factor loadings < 0.60 , one Th ‘being “super-alert”, watchful, or on guard’ and one AD-hyperactivation – ‘taking a long time to calm down’. This is explained by the fact that the sample that may be the cause of the bias. Researchers (Cyr et al., 2022) reported that items related with affective dysregulation difficulties, AD, are both hyper-activation (heightened emotional reactivity) and hypo-activation (numbness and dissociation) in the range between $< 0.6 > 0.4$. The lower internal consistency for DSO’s ‘affective dysregulation’ subscale is explained by its measure of two related but distinct constructs: hyper-activation with emotional reactivity and difficulties with calming down, and hypo-activation with chronic emotional numbing and avoidance as can both be symptoms of trauma survival (Cloitre, Shevlin, et al., 2018). Researchers (Cyr et al., 2022) theoretically assumed that both items would correlate, but failed to demonstrate strong internal consistency, thus representing the clinical reality of trauma survivors. These results are in line with previous studies which have also found similarly low internal consistency for hyperactivation subscale for the 12-item ITQ (Hyland, Shevlin, Brewin, et al., 2017) or considered hypo – and hyper-activation as separate first-order factors rather than second-order factors (Redican et al., 2021). This may also explain results that were found in the current study, especially that AD-hyperactivation item loading is relatively low, but still acceptable in both samples. In the case of the Re-experiencing item ‘having upsetting dreams’, the lower loading in the clinical sample than in the non-clinical one can be explained by the pharmacological treatment of the sleep-related symptoms during a hospital stay. Thus, patients may experience fewer nightmares (Brewin et al., 2009). Nonetheless, in our study both the Re and AD factor loadings are higher than 0.40, so it may be assumed that they are stable. Psychometric evaluations indicate that the PTSD ($\alpha = 0.810$) and DSO ($\alpha = 0.875$) subscales of the ITQ produce scores with high internal consistency, as measured by Cronbach’s alpha (α), as well as total ITQ ($\alpha = 0.887$).

The latent class analysis (LCA) results supported a four-class model, which confirmed the construct validity of ICD-11 cPTSD. The classes identified were baseline, complex PTSD, PTSD, and DSO. Similar to results reported by Kazlauskas et al. (2020), the individuals in the cPTSD class are defined by elevated PTSD and DSO symptoms with the highest AD (affect dysregulation) symptoms. In adolescents (Kazlauskas et al., 2020) the level of NSC and DR are slightly lower than they are in both our samples. There is also similarity in the PTSD domain in this study and that by Kazlauskas et al. (2020), where PTSD symptoms were elevated, and DSO symptoms are lower with still quite-high AD symptoms in both

cases. The next two domains are different when it comes to the elevation of symptoms. In adolescents the highest elevation in NSC is in the DSO domain, which is even higher than that in cPTSD. In the current study the highest elevation of symptoms is in the cPTSD domain. In the DSO domain in our study the AD symptoms are the most elevated and the NSC ones are the lowest, which is opposite to the findings from the adolescent sample (Kazlauskas et al., 2020). Those discrepancies can be attributed to the different characteristics of the two samples, but this should be verified in future studies.

Convergent and discriminant validity were evidenced by significant positive associations between ITQ symptom clusters and the DSM-5 PTSD criteria (assessed by the PCL-5 questionnaire), anxiety (measured by the PSWQ) and depression (measured by the CESD-R). Our results provide support for the concurrent, discriminant, and factorial validity of the ITQ scores, similar to previous studies with clinical and non-clinical samples (Cloitre et al., 2013; Redican et al., 2021).

The limitations of our study should be delineated. There are no other validated screening tools for cPTSD that we could use to correlate ITQ scores. The use of PCL-5 allowed us to address this issue, but not to a full extent. The DSM-5 criteria for PTSD include three that are similar to the ICD-11, but negative changes in cognition and emotions only partially overlap with the DSO symptoms (PCL-5, 2013). The selection of the two samples was not random but was actually a convenience sampling. Therefore, the results obtained do not allow for their generalizability. Information about the study was disseminated to Polish internet users aged 18–60, interested in psychology, personal development, the psychology of human development, quality of life indicators and trauma, which could in itself be a source of a bias. Although the sample size was sufficient in statistical terms, the clinical sample was relatively small and recruited only from two psychiatric hospitals based in the capital city. The potential bias of online participants declaring no psychiatric hospitalisation should also be noted. Also, despite the congruence of the findings with previous ones regarding the latent classes of ITQ, it should be noted that the goodness-of-fit indices do not provide an unequivocal answer about the best LCA model. Thus, the constructed latent class analysis should be verified in future studies.

10. Conclusion

The amended Polish adaptation of the International Trauma Questionnaire demonstrated good psychometric properties in clinical and non-clinical samples. The Polish version of the ITQ is clinically useful in

differentiating and revealing probable ICD-11 PTSD or cPTSD as part of a clinical interview.

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Data availability statement

The data that support the findings of this study are available from the corresponding author, upon reasonable request, in compliance with Polish and EU law.

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